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Imaging catheter and imaging device

Abstract

An imaging catheter and an imaging device relate to a technical field of medical devices. The imaging catheter includes a metal wire tube, a tip and a retaining ring. A metal wire exposed at one end of the metal wire tube forms a conductive end, which is electrically connected with the tip so as to allow the metal wire to be used as a conductive medium to lead static electricity out of the tip, thereby reducing effects of static electricity on a photographing assembly at the tip, and the harm of static electricity to human body. The retaining ring covers the conductive end and is connected with the tip, so as to make the metal wire tube fixedly connected with the tip.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2021/075596, filed on Feb. 5, 2021, which claims priority to Chinese Patent Application No. 2020101444005, filed on Mar. 4, 2020. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD
(1) The present disclosure relates to the technical field of medical instruments, and in particular, to an imaging catheter and an imaging device.
BACKGROUND
(2) During treatment, in order to better know about the situation inside the human body, a photographing assembly at a distal end of the imaging device can be extended into the human body, so as to image the internal condition of the human body on a screen for checking. For example, the photographing assembly that comes with a choledochoscope system can be used to directly observe pathological changes and stones in a biliary tract, which greatly improves the accuracy of the treatment of biliary diseases.
(3) In the case that the imaging device works with the photographing assembly, since the photographing assembly is an electronic component, which is highly susceptible to static electricity, the problem of unstable picture occurs, thus affecting the imaging result. At present, in order to improve the stability of picture display of the photographing assembly, for some imaging devices, a ground lead is provided at the photographing assembly of the imaging device to lead out static electricity. However, the newly added ground lead occupies limited space and leads to a complicated structure of the imaging device.
SUMMARY
(4) A purpose of the present disclosure includes, for example, providing an imaging catheter, which is capable of saving space while leading out static electricity and/or reducing the effects of static electricity on an imaged picture.
(5) The purpose of the present disclosure further includes, providing an imaging device, which is capable of saving space while leading out static electricity and/or reducing the effects of static electricity on an imaged picture.
(6) Embodiments of the present disclosure may be implemented as follows.
(7) The embodiments of the present disclosure provide an imaging catheter, including a metal wire tube, a tip and a retaining ring; a metal wire exposed at one end of the metal wire tube form a conductive end; and the retaining ring wraps the conductive end and is connected with the tip, so as to make the conductive end electrically connected with the tip.
(8) Optionally, the conductive end is electrically connected with the tip through the retaining ring.
(9) Optionally, a distal end of the conductive end includes a plurality of metal wire tips arranged in sequence along a circumferential direction of the metal wire tube, and the plurality of metal wire tips are all electrically connected with the retaining ring.
(10) Optionally, the conductive end is in contact with the tip, so as to make the conductive end

electrically connected with the tip.
(11) Optionally, the metal wire tube and the tip are arranged side by side, and the retaining ring is sleeved at a proximal end of the tip.
(12) Optionally, an outer wall of the tip includes a first peripheral surface and a second peripheral surface that are arranged in sequence along an axial direction of the tip, and a radial dimension of the first peripheral surface is smaller than that of the second peripheral surface; and
(13) an inner wall of the retaining ring is matched with the first peripheral surface.
(14) Optionally, the outer wall further includes a stepped surface, two ends of the stepped surface are respectively connected with the first peripheral surface and the second peripheral surface, and the stepped surface is configured to limit a distal end of the retaining ring.
(15) Optionally, the surface of step is disposed along a radial direction of the tip.
(16) Optionally, the metal wire tube includes an insulating tube and a metal wire; part of the metal wire is located inside the insulating tube and part of the metal wire protrudes beyond a distal end of the insulating tube to form the conductive end.
(17) Optionally, the insulating tube is thermoformed on the metal wire, so as to cover part of the metal wire.
(18) Optionally, the insulating tube is sleeved on a proximal end of the retaining ring.
(19) Optionally, the imaging catheter further includes a multi-cavity tube, which is sleeved with the metal wire tube.
(20) Optionally, the multi-cavity tube has a one-piece structure.
(21) Optionally, the metal wire tube is further provided with an export end, the export end being electrically connected with the conductive end and configured to be grounded.
(22) Optionally, the metal wire tube further includes a lead, one end of the lead being electrically connected with the conductive end and the other end of the lead being configured to be grounded so as to form the export end.
(23) An embodiment of the present disclosure further provides an imaging device, including a photographing assembly and any of the above imaging catheters, the photographing assembly being mounted in the tip of the imaging catheter.
(24) The imaging catheter and imaging device of the embodiments of the present disclosure include the following beneficial effects, for example.
(25) Embodiments of the present disclosure provide an imaging catheter, including a metal wire tube, a tip, and a retaining ring. A metal wire exposed at one end of the metal wire tube forms a conductive end, which is electrically connected with the tip, so that the metal wire may be used as a conductive medium to lead the static electricity out from the tip, thereby reducing the effects of static electricity on the photographing assembly at the tip, and reducing the harm of static electricity to the human body. At the same time, since it is not necessary to additionally arrange a ground lead at the photographing assembly, the space is effectively saved. The retaining ring covers the conductive end and is connected to the tip, so as to make the metal wire tube fixedly connected with the tip. In addition, since the conductive end is an exposed metal wire, covering of the conductive end by the retaining ring can effectively avoid the human body damage caused by a sharp part formed by the metal wire, thereby having a good effect for use.
(26) Embodiments of the present disclosure further provide an imaging device including the above imaging catheter. Since the imaging device includes the imaging catheter, it also has the beneficial effects of reducing the effects of static electricity on the photographing assembly at the tip while having a good effect for use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to illustrate the technical solutions in the embodiments of the present disclosure more clearly, the accompanying drawings that are needed in the description of the embodiments will be introduced briefly below. It should be understood that the following drawings only illustrate some embodiments of the present disclosure and cannot be considered as a limitation on its scope. For those of ordinary skill in the art, other related drawings may be obtained according to these drawings without paying a creative labor.

(2) FIG. **1** is a schematic diagram of an overall structure of an imaging device provided by an embodiment of the present disclosure;

(3) FIG. **2** is a schematic cross-sectional diagram of a partial structure of an imaging device provided by an embodiment of the present disclosure from a first perspective;

(4) FIG. **3** is a schematic structural diagram of a metal wire tube in an imaging device provided by an embodiment of the present disclosure;

(5) FIG. **4** is an enlarged schematic diagram of a partial structure at IV in FIG. **2**;

(6) FIG. **5** is a cross-sectional diagram taken along line V-V in FIG. **3**;

(7) FIG. **6** is an enlarged schematic diagram of a partial structure at VI in FIG. **2**;

(8) FIG. **7** is a schematic cross-sectional structural diagram of an imaging catheter provided by an embodiment of the present disclosure from a second perspective; and

(9) FIG. **8** is a schematic structural diagram of a tip in an imaging catheter provided by an embodiment of the present disclosure.

REFERENCE NUMERALS

(10) **10**, imaging device; **100**, imaging catheter; **110**, metal wire tube; **111**, metal mesh; **112**, conductive end; **113**, insulating tube; **114**, lead; **115**, export end; **120**, multi-cavity tube; **121**, photographing channel; **122**, working channel; **123**, leading wire channel; **124**, injection channel; **130**, tip; **131**, first peripheral surface; **132**, stepped surface; **133**, second peripheral surface; **134**, photographing hole; **135**, working hole; **136**, leading wire hole; **137**, injection hole; **140**, retaining ring; **200**, photographing assembly; **210**, lens; **220**, transmission line; **300**, handle.

DESCRIPTION OF EMBODIMENTS

(11) In order to make the purpose, technical solutions and advantages of the embodiments of the present disclosure clearer, the technical solutions in the embodiments of the present disclosure will be clearly and completely described below in conjunction with the accompanying drawings in the embodiments of the present disclosure. Apparently, the described embodiments are a part of the embodiments of the present disclosure, rather than all of the embodiments. In general, the components of the embodiments of the present disclosure described and illustrated in the drawings herein may be arranged and designed in a variety of different configurations.

(12) Therefore, the following detailed description of the embodiments of the present disclosure provided in the accompanying drawings is not intended to limit the scope of the disclosure as claimed, but is merely representative of selected embodiments of the present disclosure. Based on the embodiments in the present disclosure, all other embodiments obtained by those of ordinary skill in the art without paying creative work shall fall within the scope of protection of the present disclosure.

(13) It should be noted that similar numerals and letters refer to similar items in the following drawings. Therefore, once some item is defined in one drawing, it is not required to be further defined and explained in subsequent drawings.

(14) In the description of the present disclosure, it should be noted that in the case that the orientation or positional relationship is indicated by the terms such as “upper”, “lower”, “inner” and/or “outer”, it is based on the orientation or positional relationship shown in the accompanying drawings or is the orientation or position relationship that is customarily placed when the product of the disclosure is used, which is only for ease of describing the present disclosure and simplifying

the description, rather than indicating or implying that the referred device or element must have a particular orientation or must be constructed and operated in a particular orientation. Therefore, it should not be construed as a limitation of the present disclosure.
(15) In addition, the terms “first” and/or “second” and the like are only used for distinguishing description, and should not be construed as indicating or implying relative importance.
(16) It should be noted that the features of the embodiments of the present disclosure may be combined with each other in the case of no conflict.
(17) FIG. **1** is a schematic diagram of an overall structure of an imaging device **10** provided by an embodiment of the present disclosure; FIG. **2** is a schematic cross-sectional diagram of a partial structure of an imaging device **10** provided by an embodiment of the present disclosure from a first perspective; FIG. **3** is a schematic structural diagram of a metal wire tube **110** in an imaging device **10** provided by an embodiment of the present disclosure. Referring to FIG. **1**, FIG. **2** and FIG. **3**, this embodiment provides an imaging catheter **100**, and accordingly, provides an imaging device **10**.
(18) The imaging device **10** includes an imaging catheter **100** and a photographing assembly **200** provided in the imaging catheter **100**.
(19) Specifically, the imaging catheter **100** includes a metal wire tube **110**, a tip **130** and a retaining ring **140**. A metal wire exposed at one end of the metal wire tube **110** forms a conductive end **112**, which is electrically connected with the tip **130**, so that the metal wire inside the metal wire tube **110** can be used as a conductive medium to lead out the static electricity at the tip **130**, thereby reducing the effects of static electricity on the photographing assembly **200** at the tip **130**, and reducing the harm of static electricity to the human body; at the same time, since a ground lead is not required to be disposed at the photographing assembly **200**, a space is effectively saved. The retaining ring **140** covers the conductive end **112** and is connected with the tip **130**, so that the metal wire tube **110** is connected and fixed to the tip **130**. Furthermore, since the conductive end **112** is a metal wire that is exposed, covering of the conductive end **112** by the retaining ring **140** may effectively prevent a sharp part formed by the metal wire from damaging the human body, having a better using effect.
(20) The photographing assembly **200** includes a transmission line **220** and a lens **210** connected with a distal end of the transmission line **220**. The lens **210** is fixed in a channel inside the tip **130**, and the transmission line **220** is arranged in the metal wire tube **110**. When in use, the lens **210** of the photographing assembly **200** is fed by the imaging catheter **100** to a location that is required to be directly observed for imaging, to identify pathological changes and ensure the accuracy of treatment. At the same time, the imaging device **10** further includes a handle **300**, which is located at one end of the metal wire tube **110** away from the tip **130**. When in use, the imaging catheter **100** and photographing assembly **200** are subjected to an operation via the handle **300**.
(21) The imaging catheter **100** provided in this embodiment is further described below.
(22) FIG. **4** is an enlarged schematic diagram of a partial structure at IV in FIG. **2**; and FIG. **5** is a schematic cross-sectional diagram at V-V in FIG. **3**. Referring to FIG. **1** to FIG. **5**, in this embodiment, the metal wire tube **110** includes a metal wire and an insulating tube **113**; part of the metal wire is located in the insulating tube **113** and part of the metal wire protrudes out of a distal end of the insulating tube **113**, so that part of the metal wire is exposed outside the distal end of the insulating tube **113** to form the conductive end **112**. The conductive end **112** is electrically connected with the tip **130**, so that the static electricity at the tip **130** may be conducted via the conductive end **112** to the metal wire that is located in the insulating tube **113**. Optionally, the insulating tube **113** is subjected to thermoplastic forming on the metal wire, so as to cover part of the metal wire.
(23) It should be noted that, in this embodiment, a “proximal end” of each component in the imaging catheter **100** refers to an end of each component that is close to the handle **300**, and a “distal end” of each component in the imaging catheter **100** refers to an end of each component

away from the handle **300**. For example, the distal end of the metal wire tube **110** is the end of the metal wire tube **110** away from the handle **300**, that is, the end of the metal wire tube **110** close to the tip **130**; and the proximal end of the tip **130** is the end of the tip **130** close to the handle **300**, that is, the end of the tip **130** close to the metal wire tube **110**.

(24) Referring to FIG. **3**, in this embodiment, the metal wire tube **110** also has an export end **115**, which is electrically connected with the conductive end **112** and is configured to be grounded, thereby leading the static electricity of the tip **130** into the conductive end **112** and then to be guided to the export end **115**, so as to form a conductive path to lead out the static electricity. Specifically, the export end **115** is electrically connected with a grounding circuit board (not shown) in the handle **300**.

(25) Optionally, the metal wire tube **110** further includes a lead **114**. The lead **114** is pre-embedded in the proximal end of the metal wire, one end of the lead **114** being electrically connected with the metal wire, and the other end being connected with the grounding circuit board of the handle **300** for grounding. That is, the end of the lead **114** away from the metal wire is the export end **115** of the metal wire tube **110**, so as to make the metal wire be used as an intermediate conductive medium to form a complete conductive path, which helps to save space and simplify a structure.

(26) It should be noted that in this embodiment, the metal wire tube **110** includes the lead **114**, and the export end **115** of the metal wire tube **110** is an end of the lead **114** that is connected to the grounding circuit board of the handle **300**. It can be understood that in other embodiments, the export end **115** may also be formed in other ways as required. For example, the insulating tube **113** at the proximal end of the metal wire tube **110** is peeled off to expose a metal mesh **111** at this position, thereby allowing the metal wire exposed at the proximal end of the metal mesh **111** to form the export end **115**.

(27) FIG. **6** is an enlarged schematic diagram of a partial structure at VI in FIG. **2**; FIG. **7** is a schematic cross-sectional structural diagram of an imaging catheter **100** provided by the embodiment from a second perspective. Referring to FIG. **2**, FIG. **6** and FIG. **7**, in this embodiment, the imaging catheter **100** further includes a multi-cavity tube **120** disposed in the metal wire tube **110**.

(28) Specifically, in the process of production and installation, first, a multi-strand metal wire is used to form a tubular metal mesh **111** by weaving outside of the multi-cavity tube **120**, and an end surface of a distal end of the tubular metal mesh **111** is flush with that of a distal end of the multi-cavity tube **120**. A single-cavity tube made of insulating material is then sleeved outside the metal mesh **111**, and is subsequently melted to coat the metal mesh **111** by hot melting, so as to form a tubular member by which the metal mesh **111** is coated after cooling, that is, the insulating tube **113** is thermoformed. At this point, the metal mesh **111** is entirely wrapped inside the insulating tube **113**; at the same time, the formed insulating tube **113** is tightly adhered to the multi-cavity tube **120** by means of hot melting, which has a robust connection and high reliably. Finally, the insulating tube **113** at the distal end of the metal mesh **111** is peeled off to make the metal wire that is located at the distal end of the metal mesh **111** exposed outside of the insulating tube **113**, so as to form the conductive end **112**, and the conductive end **112** is electrically connected with the tip **130** to lead out the static electricity at the tip **130**. It can be understood that, in other embodiments, other manufacturing methods may also be used; for example, a metal wire tube **110** is prefabricated, the multi-cavity tube **120** is then allowed to penetrate through the metal wire tube **110**, and finally, the metal wire tube **110** is connected with the multi-cavity tube **120** by means of bonding or the like. Moreover, in this embodiment, multi-strand metal wire is woven to form a mesh. It can be understood that in some other embodiments, it is also possible to arrange the multi-strand metal wire at intervals along a circumferential direction of the multi-cavity tube **120** as required.

(29) It should be noted that in this embodiment, the conductive end **112** is formed by peeling off the insulating tube **113** at the distal end of the metal mesh **111** to expose the metal wire. It can be understood that in this embodiment, the conductive end **112** may also be formed in other ways as

required. For example, a single-cavity tube with a length smaller than the metal mesh **111** is selected to be sleeved outside the metal mesh **111**, so that one end of the metal mesh **111** protrudes out of the single-cavity tube and is then exposed outside the single-cavity tube; after hot-melting of the single-cavity tube, the metal wire that is located at one end of the metal mesh **111** and exposed outside the single-cavity tube naturally protrudes out of the insulating tube **113** formed, thereby forming the conductive end **112**. At the same time, it may also be understood that in other embodiments, the insulating tube **113** may also be formed by other ways, such as jetting and extrusion.
(30) By arranging the metal wire tube **110** outside the multi-cavity tube **120**, the strength of the multi-cavity tube **120** may be effectively enhanced, so that the imaging catheter **100** has a good pushing performance, and in case of passing through a curved channel, the damage of the multi-cavity tube **120** is obviated and the degree of deformation of the cross section of the multi-cavity tube **120** may be reduced. In other words, when the multi-cavity tube **120** passes through the curved channel, the shape of each cross section at a bend may still remain substantially circular; therefore, it may be ensured that channels in the multi-cavity tube **120** are not deformed. It should be noted that in this embodiment, the “cross section” refers to a plane which is cut from a plane perpendicular to the axis of the multi-cavity tube **120**.
(31) Optionally, the multi-cavity tube **120** has a one-piece structure, that is, the multi-cavity tube **120** has a complete one-piece tubular structure, which may be integrally molded by injection molding or pouring, etc., or be integrally formed by bonding or welding, etc. It may be understood that in other embodiments, the multi-cavity tube **120** may also be set to be formed by splicing multiple materials as required.
(32) Referring to FIG. **5** and FIG. **7**, the multi-cavity tube **120** has a plurality of channels, which are divided into a photographing channel **121**, a working channel **122**, a leading wire channel **123** and an injection channel **124**, according to respective functions.
(33) Specifically, many channels in the multi-cavity tube **120** are all circular channels. The photographing channel **121** and the working channel **122** are arranged at intervals in the middle of the multi-cavity tube **120** along a radial direction of the multi-cavity tube **120**; that is, the connection line formed by the center of the photographing channel **121** and the center of the working channel **122** passes through the center of the multi-cavity tube **120**, where the connection line is a first connection line L**1**. The radial dimension of the photographing cavity **121** is smaller than that of the working channel **122**. There are two injection channels **124**, which are respectively located on both sides of the connection line L**1**, and the centers of the two injection channels **124** are connected to form a connection line that is a second connection line L**2**. At the same time, in order to facilitate the arrangement of the two injection channels **124**, ensure the injection volume and facilitate the injection operation, the intersection between the second connection line L**2** and the first connection line L**1** is located between the photographing channel **121** and the working channel **122**. There are four leading wire channels **123**, which are oppositely arranged on both side of the first connection line L**1** in a group of two. That is, two of the four leading wire channels **123** are located on one side of the first connection line L**1**, and the other two are located on the other side of the first connection line L**1**. At the same time, the two leading wire channels **123** that are located on the same side of the first connection line L**1** are distributed on two sides of the second connection line L**2**, respectively.
(34) It should be noted that there is no limitation on the number, location and/or size of the channel in the multi-cavity tube **120** herein. It can be understood that in other embodiments, the channel may also be provided as required, for example, the number of the leading wire channels **123** may be set to two, etc.
(35) FIG. **8** is a schematic structural diagram of a tip **130** in an imaging catheter **100** provided by this embodiment. Referring to FIG. **2** and FIG. **8**, the tip **130** of the imaging catheter **100** is located at the distal end of the imaging catheter **100**. In this embodiment, the tip **130** is made of medical

grade metal materials. Optionally, the material of the tip **130** is medical grade 304-type stainless steel. It can be understood that in some other embodiments, the tip **130** may also be made of other medical grade metal materials or rigid plastic materials as required, such as PEEK (polyetheretherketone).
(36) Referring to FIG. **5**, FIG. **7** and FIG. **8**, the tip **130** is provided with a plurality of through-holes, which are each arranged to penetrate through the tip **130** along an axial direction of the tip **130**. These through-holes are provided in one-to-one correspondence with the plurality of channels in the multi-cavity tube **120**. Correspondingly, the plurality of through-holes at the tip **130** are respectively divided into an injection hole **137**, a leading wire hole **136**, a working hole **135** and a photographing hole **134** according to respective functions.
(37) The injection hole **137** is provided to correspond to the injection channel **124**; that is, the injection hole **137** is communicated with the injection channel **124**, so that liquid can be injected through the injection channel **124** and the injection hole **137**. For example, during the imaging process of the photographing assembly **200**, if the lens **210** of the photographing assembly **200** is blocked due to bleeding, physiological saline or the like may be injected through the injection channel **124** and the injection hole **137** to flush the lens **210**. Specifically, the number of injection holes **137** is set to two, and two injection holes **137** are provided in one-to-one correspondence with the two injection channels **124**.
(38) The leading wire hole **136** is arranged to correspond to the leading wire channel **123**; that is, the leading wire channel **123** is communicated with the leading wire hole **136** for allow the leading wire to pass through. Specifically, the number of the leading wire hole **136** is set to four, and four leading wire holes **136** are provided in one-to-one correspondence with four leading wire channels **123**.
(39) The working hole **135** is arranged to correspond to the working channel **122**; that is, the working channel **122** is communicated with the working hole **135** to meet the operation and treatment requirements. For example, some operation instruments are inserted into the human body through the working channel **122** and the working hole **135**.
(40) The photographing hole **134** corresponds to the photographing channel **121**; that is, the photographing hole **134** is communicated with the photographing channel **121**. The lens **210** of the photographing assembly **200** is located in the photographing hole **134**, and acquires human body images through the opening of the photographing hole **134** which is disposed at the distal end of the tip **130**. The transmission line **220** of the photographing assembly **200** is arranged to pass through the photographing channel **121**, the distal end of the transmission line **220** is connected with the lens **210**, and the proximal end of the transmission line **220** extends to the handle **300** and is connected with a display device (not shown) to output an imaged picture.
(41) It should be noted that the number, location, size and function of the plurality of through-holes in the tip **130** are not limited herein. It can be understood that in other embodiments, it may also be set as required. For example, only some of the injection holes **137**, the leading wire holes **136** and the working holes **135** are selected to be disposed in the tip **130**, as required; or if there are two leading wire channels **123** of the multi-cavity tube **120**, the number of leading wire holes **136** is accordingly set to two.
(42) Referring to FIG. **1** to FIG. **4**, in this embodiment, the imaging catheter **100** further includes a retaining ring **140**, which is sleeved on the conductive end **112** and connected with the tip **130**, so as to fix the metal wire tube **110** with the tip **130**. Moreover, the retaining ring **140** wraps the end of the conductive end **112**, so that a sharp structure formed by the bare metal wire end which forms the conductive end **112** is wrapped in the retaining ring **140**, thereby preventing the human body from being damaged by such sharp structure.
(43) Along the axial direction of the imaging catheter **100**, the metal wire tube **110** is arranged side by side with the tip **130**; that is, the end face of the distal end of the metal wire tube **110** and the end face of the proximal end of the tip **130** are matched with each other in a butted manner or are

arranged at a small distance apart; moreover, in the case that the end face of the distal end of the metal wire catheter **110** and the end face of the proximal end of the tip **130** are arranged at intervals, the metal wire tube **110** and the tip **130** are arranged at intervals. That is, there is no nesting relationship between the metal wire tube **110** and the tip **130**. The proximal end of the retaining ring **140** is sleeved on the metal wire tube **110**, and the distal end of the retaining ring **140** is sleeved on the tip **130**, so that the metal wire tube **110** and the tip **130** are fixed to each other.
(44) Further, the retaining ring **140** completely wraps the conductive end **112**, and the proximal end of the retaining ring **140** is extended and sleeved outside the insulating tube **113**, so the wire exposed outside the insulating tube **113** are completely covered by the retaining ring **140**, which prevents the wire from being exposed and affecting its use.
(45) In this embodiment, since the end face of the distal end of the metal mesh **111** formed by the wire is flush with that of the distal end of the multi-cavity tube **120** and there is no nesting relationship between the metal wire tube **110** and the tip **130**, the retaining ring **140** is provided with a conductive portion, so as to electrically connect the conductive end **112** and the tip **130**. At the same time, the conductive portion is electrically connected with each of the conductive end **112** and the tip **130**, thereby realizing the electrical connection between the conductive end **112** and the tip **130** via the conductive portion. That is, the conductive end **112** is electrically connected with the tip **130** through the retaining ring **140**.
(46) It should be noted that in this embodiment, the retaining ring **140** is used as an intermediate conductive medium between the conductive end **112** and the tip **130**, and the conductive end **112** is electrically connected with the tip **130** through the retaining ring **140**. It can be understood that, in some other embodiments, the electrical connection of the conductive end **112** with the tip **130** may be realized through contact between the conductive end **112** and the tip **130**, as required. For example, the distal end of the metal mesh **111** is configured to protrude beyond the multi-cavity tube **120** in the axial direction, so that the conductive end **112** disposed at the distal end of the metal mesh **111** is in contact with the tip **130**; alternatively, the proximal end of the tip **130** is sleeved on the conductive end **112**, so that an inner peripheral surface of the proximal end of the tip **130** is in contact with the conductive end **112**, so as to realize the electrical connection between the conductive end **112** and the tip **130**.
(47) Since in the metal wire tube **110**, multi-strand metal wires are woven to form a tubular metal mesh **111**, the conductive end **112** is a part that is at the distal end of the metal mesh **111** and is exposed outside the insulating tube **113**, and so the conductive end **112** is annular. A plurality of metal wire tips are arranged at the distal end of the conductive end **112** along the circumferential direction of the metal wire tube **110**, the metal wire tip being an end formed by cutting off the metal wire. The conductive portion in the retaining ring **140** is annular and the plurality of metal wire tips are each electrically connected with the conductive portion, so that all the metal wires may export static electricity from the conductive portion. That is, all metal wires may be used for grounding to lead out the static electricity, so that the grounding effect can be effectively improved and the leading out effect of the static electricity is better.
(48) Optionally, the retaining ring **140** is made of metal material; that is, the whole retaining ring **140** is conductive. In other words, the whole retaining ring **140** may be used as a conductive part to make the conductive end **112** be electrically connected with the tip **130**. It can be understood that the specific material of the retaining ring **140** is not limited in this embodiment, as long as it may be used as an intermediate conductive medium to realize electrical connection of the conductive end **112** with the tip **130**.
(49) It should be noted that, in this embodiment, the retaining ring **140** is made of metal material, so that the whole retaining ring **140** is conductive. It can be understood that, in some other embodiments, as required, only part of the retaining ring **140** is made of metal material that is conductive. For example, a metal coating is provided at a place where the inner wall of the retaining ring is in contact with the tip **130** and the conductive end **112**, so as to electrically connect

the conductive end **112** and the tip **130**; and an annular conductive portion may be formed, as long as all the metal wires may be electrically connected with the tip **130**.
(50) Referring to FIG. **2**, FIG. **6** and FIG. **8**, in this embodiment, along the axial direction of the tip **130**, the outer wall of the tip **130** includes a first peripheral surface **131** and a second peripheral surface **133** that are arranged in sequence, and a radial dimension of the first peripheral surface **131** is smaller than that of the second peripheral surface **133**. Specifically, the first peripheral surface **131** is located at the proximal end of the tip **130**, and the second peripheral surface **133** is located at the distal end of the tip **130**. The inner wall of the retaining ring **140** is matched with the first peripheral surface **131**, so that the radial dimension of a connection between the retaining ring **140** and the tip **130** may be effectively reduced, which is convenient to operate the imaging catheter **100** into the human body.
(51) The first peripheral surface **131** is cylindrical surface, and the proximal end of the second peripheral surface **133** is also cylindrical. Meanwhile, the radial dimension of the first peripheral surface **131** is smaller than that of the proximal end of the second peripheral surface **133**, so that an outer portion of the tip **130** forms a stepped structure. The tip **130** also includes a stepped surface **132**, which is located between the first peripheral surface **131** and the second peripheral surface **133**, and two ends of the stepped surface **132** are respectively connected with the first peripheral surface **131** and the second peripheral surface **133**. Since the radial dimension of the first peripheral surface **131** is smaller than that of the second peripheral surface **133**, the first peripheral surface **131** and the stepped surface **132** are connected to form a notch recessed inward in a radial direction, relative to the second peripheral surface **133**, and the distal end of the retaining ring **140** is arranged in this notch. In this embodiment, the retaining ring **140** is connected with the tip **130** by laser welding. Optionally, in some other embodiments, the retaining ring **140** may be connected with the tip **130** in other ways such as gluing; or the distal end of the retaining ring **140** may be pressed against the first peripheral surface **131**.
(52) In this embodiment, the notch is formed by performing machining at the cylindrical proximal end of the tip **130**. Optionally, in some other embodiments, the tip **130** may also be formed in other ways, for example, it may be integrally formed by powder metallurgy.
(53) At the same time, the end face of the distal end of the retaining ring **140** abuts against the stepped surface **132**, so that the retaining ring **140** is limited by the stepped surface **132**, so as to ensure that the retaining ring **140** is connected with the tip **130** reliably. Optionally, the surface of step **132** is arranged along a radial direction of the tip **130**; that is, the plane where the surface of step **132** is located is perpendicular to the axis of the tip **130**, and at the same time, the stepped surface **132** is disposed perpendicular to both the first peripheral surface **131** and the second peripheral surface **133**.
(54) According to an imaging catheter **100** provided in this embodiment, the production process and working principle of the imaging catheter **100** are as follows.
(55) During installation and production, a tubular metal mesh **111** is firstly formed, outside a multi-cavity tube **120**, by weaving multi-strand wires; then a single-cavity tube made of insulating material is sleeved outside the metal mesh **111**, and at the same time, a lead **114** is pre-embedded and electrically connected with the distal end of the metal mesh **111**. Next, the single-cavity catheter is melted by means of hot-melting and then solidified to thermoform an insulating tube **113**. The metal mesh **111** is wrapped by the insulating tube **113**, so as to form a metal wire tube **110**. The portion of insulating tube **113** disposed at the distal end of the metal wire tube **110** is then peeled off, so that a metal wire forming the distal end of the metal mesh **111** leaks out of the insulating tube **113**, so as to form a conductive end **112**. Finally, two ends of a retaining ring **140** are sleeved on the metal wire tube **110** and a tip **130**, respectively, and the retaining ring **140** is fixedly connected with the tip **130** by laser welding, so that the metal wire tube **110** is fixed connected with the tip **130** by the retaining ring **140**. Meantime, the conductive end **112** is covered by the retaining ring **140**, which prevents the wire exposed from affecting use and causing damage.

(56) When in use, a photographing assembly **200** is firstly put into an imaging catheter **100**, a lens **210** of the photographing assembly **200** is held in a photographing hole **134** of the tip **130**, and a transmission line **220** of the photographing assembly **200** extends, along a photographing channel **121** of the multi-cavity tube **120**, to a handle **300** and is connected with a display device. The imaging catheter **100** equipped with the photographing assembly **200** is then extended into the human body through an endoscopic channel to the desired observation position, so that the desired picture is acquired by the photographing assembly **200** and is imaged on the display device. In the process of observation and use, the static electricity at the tip **130** is led out, through a conductive path formed by connecting the retaining ring **140**, the conductive end **112**, the metal wire in the insulating tube **113**, and the lead **114** in sequence, into the handle **300** for grounding, so as to lead out the static electricity, thereby preventing the operation of the camera component **200** from being affected by static electricity.
(57) The imaging catheter **100** provided by this embodiment has at least the following advantages.
(58) Embodiments of the present disclosure provide an imaging catheter **100**, which uses the metal wire wrapped outside the multi-cavity tube **120** as a conductive medium to lead out the static electricity, so as to prevent the imaged picture from being affected by static electricity, to reduce the harm of static electricity to the human body; at the same time, to effectively save the space of the imaging catheter **100**, as a result, the overall structure of the imaging catheter **100** is less affected. The metal wire tube **110** is connected with the tip **130** by the retaining ring **140**, the metal wire exposed may be wrapped with the annular retaining ring **140**, so as to prevent the human body injury caused by a sharp structure formed by the wire in use. At the same time, the conductive end **112** is electrically connected with the tip **130** through the retaining ring **140**, so that all metal wires forming the conductive end **112** may be used for grounding, with a good effect of grounding.
(59) This embodiment also provides an imaging device **10** including the above imaging catheter **100**. Since the imaging device **10** includes the above imaging catheter **100**, the imaging device **10** also has the beneficial effects of preventing the imaged picture from being affected by static electricity, saving space, avoiding damage to the human body, and having a good effect of grounding.
(60) The above are merely specific implementations of the present disclosure, but the scope of protection of the present disclosure is not limited thereto. Any modification or substitution that can be easily conceived by those skilled in the art within the technical scope disclosed in the present disclosure should be covered by the scope of protection of the present disclosure. Therefore, the scope of protection of the present disclosure should be subject to the scope of protection of the claims.

Claims

1. An imaging catheter, comprising: a metal wire tube, a tip and a retaining ring; wherein a metal wire is exposed at one end of the metal wire tube to form a conductive end; and the retaining ring wrapping the conductive end is connected to the tip to make the conductive end electrically connected with the tip; the metal wire tube has an export end electrically connected with the conductive end and configured to be grounded, wherein the metal wire tube further comprises a lead, one end of the lead being electrically connected with the conductive end and the other end of the lead being configured to be grounded to form the export end.
2. The imaging catheter according to claim 1, wherein the conductive end is electrically connected with the tip through the retaining ring.
3. The imaging catheter according to claim 2, wherein a distal end of the conductive end comprises a plurality of metal wire tips arranged in sequence along a circumferential direction of the metal wire tube, and the plurality of metal wire tips are all electrically connected with the retaining ring.
4. The imaging catheter according to claim 1, wherein the conductive end is in contact with the tip,

so as to make the conductive end electrically connected with the tip.
5. The imaging catheter according to claim 1, wherein the metal wire tube and the tip are arranged side by side, and the retaining ring is sleeved at a proximal end of the tip.
6. The imaging catheter according to claim 5, wherein an outer wall of the tip comprises a first peripheral surface and a second peripheral surface that are arranged in sequence along an axial direction of the tip, and a radial dimension of the first peripheral surface is smaller than that of the second peripheral surface; and an inner wall of the retaining ring is matched with the first peripheral surface.
7. The imaging catheter according to claim 6, wherein the outer wall further comprises a stepped surface, two ends of the stepped surface are respectively connected with the first peripheral surface and the second peripheral surface, and the stepped surface is configured to limit a distal end of the retaining ring.
8. The imaging catheter according to claim 7, wherein the stepped surface is disposed along a radial direction of the tip.
9. The imaging catheter according to claim 1, wherein the metal wire tube comprises an insulating tube and the metal wire; part of the metal wire is located inside the insulating tube and part of the metal wire protrudes beyond a distal end of the insulating tube to form the conductive end.
10. The imaging catheter according to claim 9, wherein the insulating tube is thermoformed on the metal wire, so as to cover part of the metal wire.
11. The imaging catheter according to claim 9, wherein the insulating tube is sleeved on a proximal end of the retaining ring.
12. The imaging catheter according to claim 1, wherein the imaging catheter further comprises a multi-cavity tube, which is sleeved with the metal wire tube.
13. The imaging catheter according to claim 12, wherein the multi-cavity tube has a one-piece structure.
14. An imaging device, wherein the imaging device comprises a photographing assembly and the imaging catheter according to claim 1, the photographing assembly being mounted in the tip of the imaging catheter.